

LOGBOEK – Game development exam (K0788)

Basis Programmeren van Games

Student: Hamilton Posada

Opleiding: Bit Academy

Ontwikkelperiode: 30 augustus 2025 – 04 October 2025

Totale ontwikkeltijd: 5 weken

GitHub: <https://github.com/HamiltonPDev/RPG-Unity-game>

Totaal commits: 55

DEVELOPMENT TIMELINE

Augustus 2025 - Week 1 (30 augustus)

Commits: 8

Tijd besteed: ~20 uur

Dag 1 - 30 augustus:

16:59 - Initial commit

17:09 - This is the project for my KD examen

17:20 - Added some gitignore

17:26 - Removed some files that doesn't need in the repository

19:25 - I have been created the spawnzone for the player follow

20:48 - The spawn zone is set up and ready to use also the dontdestroyonload

20:51 - I change the start form the awake because its more stable

21:21 - Updated the scripts for the go to new place exactly where the player must go

Geïmplementeerd:

- Project setup (Unity 6000.1.10f1)
- Git configuratie + .gitignore
- `SpawnZone.cs` - Player positioning systeem
- `DontDestroyOnLoad.cs` - Object persistence
- `GoToNewPlace.cs` - Scene transitions

Reflectie dag 1:

- Productieve eerste dag met 8 commits
 - Focus op scene management foundations
 - Goede basis gelegd voor multi-scene game
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September 2025 - Week 1 (1-3 september)

Commits: 9

Tijd besteed: ~25 uur

Dag 2-4 (1-3 september):

1 sep 12:21 - I have added a new scene in the project
1 sep 12:31 - Enemy character added
1 sep 14:31 - The enemy character updated
1 sep 15:25 - feat: Add enemy prefab and damage player script
1 sep 15:46 - feat: Implement health management and damage system for player
2 sep 12:19 - feat: Update player sprite metadata and add weapon damage script
2 sep 14:14 - feat: Add health management to enemy prefab with max health property
2 sep 14:18 - feat: Update weapon damage logic to apply damage to enemies
3 sep 12:02 - Add new tag 'Middleground'
3 sep 12:17 - feat: Add Player_Attacking_Down animation

Geïmplementeerd:

- Nieuwe scene toegevoegd
- Enemy character + prefab
- `EnemyController.cs` - Random patrol
- `DamagePlayer.cs` - Player collision damage
- `HealthManager.cs` - Health systeem
- `WeaponDamage.cs` - Attack damage
- Player attack animation (down)
- Tag systeem

Problemen & oplossingen:

- **Probleem:** Weapon destroyde enemies direct
- **Oplossing:** Changed to damage system met health

September 2025 - Week 2 (8-11 september)

Commits: 6

Tijd besteed: ~18 uur

8 sep 14:24 - Add player attack animations (left, right, up)
9 sep 15:22 - Enhance player animations and controller logic
10 sep 10:19 - Implement code changes to enhance functionality
10 sep 10:26 - fix: Update PolygonCollider2D to be a trigger
10 sep 16:26 - Add Blood Burst prefab

11 sep 12:14 - Add EmojiOne sprite sheet for TextMesh Pro

11 sep 12:25 - fix: Update DamageNumber script

Geïmplementeerd:

- Attack animations alle 4 richtingen
 - Animator state transitions
 - `DamageNumber.cs` - Floating damage
 - Blood burst visual effect
 - TextMeshPro setup
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September 2025 - Week 3 (15-20 september)

Commits: 8

Tijd besteed: ~24 uur

15 sep 12:21 - feat: Enhance DamageNumber functionality and add UIManager

15 sep 12:38 - Fix: change maxHealth and currentHealth to public

15 sep 14:30 - feat: update prefabs and enhance damage handling

15 sep 16:00 - Add health flashing effect and create CharacterStats script

15 sep 16:26 - feat: Add experience handling for defeated enemies

17 sep 11:47 - feat: Organize UIManager script

17 sep 14:09 - feat: Enhance character stats with level-based calculations

18 sep 13:13 - feat: update enemy and player stats, enhance UI

19 sep 14:02 - feat: remove damage property from prefab

20 sep 13:19 - added shadow single.png

Geïmplementeerd:

- `UIManager.cs` - Real-time display
 - `CharacterStats.cs` - Level/XP/Stats systeem
 - I-frame flashing effect
 - Experience system (XP on kill)
 - Level progression
 - Stat scaling
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September 2025 - Week 4 (22-26 september)

Commits: 11

Tijd besteed: ~28 uur

22 sep 11:33 - feat: add EnemyStats script

22 sep 12:24 - feat: update enemy stats with level-based scaling

22 sep 14:50 - feat: update EnemyStats with player stats reference
22 sep 16:04 - feat: refactor EnemyStats
22 sep 16:51 - feat: update EnemyStats and DamagePlayer for health scaling
23 sep 11:57 - fix: correct health scaling using original base health
23 sep 12:33 - feat: enhance uimanager
25 sep 12:00 - Add fantasy icon pack assets
25 sep 13:50 - fix update damage calculation in DamagePlayer
25 sep 14:31 - feat: add player defense display
25 sep 16:44 - Refactor player attack timing and add enemy animations
26 sep 10:59 - fix: enable circle collider 2d for enemy prefab

Geïmplementeerd:

- `EnemyStats.cs` - **KERN FEATURE**
 - Dynamic enemy scaling
 - Health + damage per level
- Defense calculation systeem
- UI voor defense
- Enemy animations
- Fantasy icons

Bug fixes:

- Health scaling met originalBaseHealth
- Negative damage prevention
- Collider fixes

September 2025 - Week 5 (29 september)

Commits: 8

Tijd besteed: ~12 uur

29 sep 09:58 - docs: update README.md
29 sep 10:18 - docs: add note about incomplete AI
29 sep 12:02 - docs: update README for structure
29 sep 12:26 - Implement code changes in readme
29 sep 12:27 - docs: fix punctuation
29 sep 12:32 - docs: standardize formatting
29 sep 13:34 - docs: update with MoSCoW backlog
29 sep 13:45 - Docs: update technical features

Gedaan:

- Volledige README documentatie

- MoSCoW status update
- Project structure docs

Here's the text formatted for easy copy-paste into Word:

SAMENVATTING

Periode: 30 augustus - 30 september 2025 (1 maand)

Commits: 66

Actieve dagen: ~20 dagen

Gemiddeld: 3 commits per actieve dag

Dit project is ontwikkeld over een periode van 1 maand met intensieve development sprints. Het examen heeft een officiële tijdsduur van 2 weken, maar extra tijd is gebruikt voor Enemy AI implementation, polish en documentatie.

HUIDIGE STATUS (30 september 2025)

VOLTOOID ✓

1. ✓ Core gameplay (Player movement, combat, health)
2. ✓ Enemy AI state machine (COMPLEET - vandaag!)
3. ✓ Stats & leveling system
4. ✓ Enemy scaling system
5. ✓ Scene management
6. ✓ UI displays
7. ✓ Visual feedback systems
8. ✓ Documentatie basis (Logboek concept, Ontwikkelomgeving)

SEPTEMBER 2025 - Dag 15 (30 september) - ENEMY AI COMPLETION DAY

Commits: 11

Tijd besteed: 8 uur

Status: MAJOR MILESTONE - Enemy AI volledig geïmplementeerd

Dit was de belangrijkste dag van het hele project. Van een lege EnemyAI.cs template naar een volledig functionele state machine met patrol, chase, attack en idle behaviors. De attack is niet volledig klaar.

Chronologische ontwikkeling:

10:11 - Feat: enhance enemyai with patrol and detection features
BASIS: Patrol movement + player detection range

10:16 - Feat: add new enemy behavior component and safety checks
VEILIGHEID: Null checks voor player reference

12:08 - Feat: implement patrol behavior and state management
STATE MACHINE: Eerste versie Patrol/Chase states

12:13 - feat: add comments to enemy ai parameters
DOCUMENTATIE: Alle parameters toegelicht

13:28 - feat: refine enemy ai with improved state management
VERFIJNING: Patrol → Chase → Attack transitions

13:49 - Enhance enemy ai with refined attack and Idle state management
UITBREIDING: Idle state toegevoegd voor waypoint waiting

13:53 - Implement state management for enemy ai
COMPLETION: Alle 4 states werkend (Patrol/Chase/Attack/Idle)

14:13 - Enhance enemy ai by adding attack conditions in animator
ANIMATIE: Attack state triggers in Animator

14:47 - visualization for attack and detection ranges
DEBUG: Gizmos voor range visualization (yellow/red circles)

15:32 - Refine enemy AI by adjusting detection and attack ranges
FINAL TUNING: Detection 5f, Attack 1.5f, speeds optimized

Technische achievement:

EnemyAI:

- State machine met 4 states
- Smooth transitions tussen behaviors
- Animator integratie
- Gizmos debugging visualization
- Auto-generated patrol points
- Safety checks overall

Reflectie einde dag:

Dit was technisch het moeilijkste onderdeel van het hele project. Gelukt om in één gefocuste dag van empty script naar production-ready AI te gaan. State pattern maakt toekomstige uitbreidingen eenvoudig.

EnemyController script is nu deprecated (marked as "learning script").